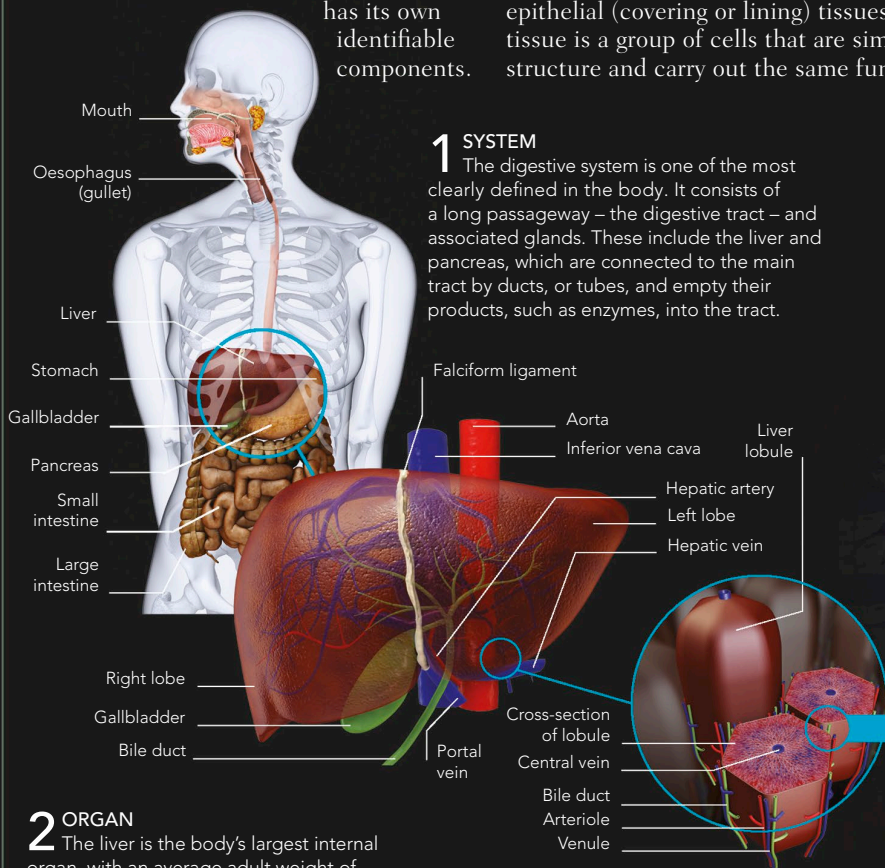


BODY SYSTEMS TO CELLS

EACH SYSTEM CAN BE SEEN AS A HIERARCHY. THE SYSTEM ITSELF IS AT THE TOP OF THE HIERARCHY; NEXT ARE ITS ORGANS; THEN THE TISSUES THAT MAKE UP THE ORGANS; AND AT THE BOTTOM ARE THE CELLS FROM WHICH TISSUES ARE MADE.

A body system is usually regarded as a collection of organs and parts designed for one important task. The systems are integrated and interdependent, but each has its own identifiable components.

The main parts of a system are its organs and tissues. Most organs are composed of different tissues. The brain, for example, contains nervous, connective, and epithelial (covering or lining) tissues. A tissue is a group of cells that are similar in structure and carry out the same function.



1 SYSTEM

The digestive system is one of the most clearly defined in the body. It consists of a long passageway – the digestive tract – and associated glands. These include the liver and pancreas, which are connected to the main tract by ducts, or tubes, and empty their products, such as enzymes, into the tract.

2 ORGAN

The liver is the body's largest internal organ, with an average adult weight of 1.5kg (3½lb), which is slightly more than the brain. Within the liver is a system of tubes for carrying away its digestive product, bile, which is stored in the small sac under its right end, the gallbladder.

3 ORGAN SUBSTRUCTURE

The structural-functional units of the liver are hepatic lobules. The lobules are six-sided and have blood vessels and bile ducts inside and between them.